

Reflection

Answer the following questions for all enhancements which did NOT lead to increased performance:

- Why did the changes not result in the increased performance?

Stopword removal reduced accuracy by 0.77% because the NLTK stopwords list includes negation words such as "not", "no", "won't", and "can't". In short tweets, these words are critical for distinguishing negative from positive sentiment. Removing them caused the model to misclassify negated positive phrases as positive.

The stemmer reduced the model accuracy by 0.05%. One possible reason is that stemming removes useful information from words by reducing them to their root forms. This process can eliminate important contextual or semantic meaning. In addition, different words may become identical after stemming, which decreases the quality of the features used for prediction. For this dataset, the original words likely contained more meaningful information than the simplified stemmed versions, leading to slightly better prediction performance without stemming.

Embeddings reduced the model accuracy by 7.40%. One possible reason is that the effective vocabulary only contained 764 words, causing many tweet tokens to return zero vectors because they were not included in the embedding vocabulary. As a result, a large amount of useful textual information was lost, which negatively affected the model's prediction performance.

- How can the performance be improved?

One possible improvement would be to use a custom stopwords list that explicitly keeps negation words such as "not," "no," and "never." These words often carry important sentiment and contextual meaning, so removing them may reduce prediction accuracy. Retaining negation words could help the model better understand the meaning of the text and improve performance.

Answer the following questions for all enhancements which DID lead to increased performance:

- What was the magnitude of the performance increase?

The largest improvement came from using Stemmer and Bigrams together to increase 0.54% over baseline (65.72% → 66.26%). Bigrams alone contributed +0.42%, making it the single most impactful enhancement. Combinations involving stopwords reduced the gains from bigrams, capping improvements between +0.11% and +0.21%.

- Other than the specific enhancements, could the performance increase be due to any other factors?

Stemming reduces the vocabulary size, causing bigrams to appear more frequently in the training data. As a result, the model is able to see more examples for each feature, which may help improve feature learning and reduce data sparsity.